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```
setwd("C:/Users/Cynthiali0621/Desktop")
library(survival)
```

```
## Warning: package 'survival' was built under R version 4.4.3
```

```
data <- read.csv("leuk.csv")
```

```
#a)
# Make a Surv object.  'type = "right"' is for right-censoring
surv_obj <- with(data, Surv(time, cens, type = "right"))

# Fit Kaplan-Meier by treatment group
km_fit <- survfit(surv_obj ~ group, data = data)

# Look at survival estimates at weeks 1 through 12
summary(km_fit, times = 1:12)
```

```
## Call: survfit(formula = surv_obj ~ group, data = data)
##
##                  group=6 MP
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     21       0    1.000  0.0000        1.000        1.000
##     2     21       0    1.000  0.0000        1.000        1.000
##     3     21       3    0.857  0.0764        0.720        1.000
##     4     17       1    0.807  0.0869        0.653        0.996
##     5     16       1    0.756  0.0950        0.591        0.967
##     6     13       0    0.756  0.0950        0.591        0.967
##     7     12       1    0.693  0.1059        0.514        0.935
##     8     11       1    0.630  0.1135        0.443        0.897
##     9     10       0    0.630  0.1135        0.443        0.897
##    10      9       0    0.630  0.1135        0.443        0.897
##    11      7       1    0.540  0.1281        0.339        0.860
##    12      6       1    0.450  0.1347        0.250        0.809
##
##                  group=Placebo
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     21       4   0.8095  0.0857      0.65785        0.996
##     2     17       3   0.6667  0.1029      0.49268        0.902
##     3     14       2   0.5714  0.1080      0.39455        0.828
##     4     12       4   0.3810  0.1060      0.22085        0.657
##     5      8       0   0.3810  0.1060      0.22085        0.657
##     6      8       4   0.1905  0.0857      0.07887        0.460
##     7      4       0   0.1905  0.0857      0.07887        0.460
##     8      4       1   0.1429  0.0764      0.05011        0.407
##     9      3       1   0.0952  0.0641      0.02549        0.356
##    10      2       0   0.0952  0.0641      0.02549        0.356
##    11      2       1   0.0476  0.0465      0.00703        0.322
##    12      1       1   0.0000     NaN           NA           NA
```

```
#b
# Print a summary of the full KM curves
summary(km_fit)
```

```
## Call: survfit(formula = surv_obj ~ group, data = data)
##
##                 group=6 MP
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     3     21       3    0.857  0.0764        0.720        1.000
##     4     17       1    0.807  0.0869        0.653        0.996
##     5     16       1    0.756  0.0950        0.591        0.967
##     7     12       1    0.693  0.1059        0.514        0.935
##     8     11       1    0.630  0.1135        0.443        0.897
##    11      7       1    0.540  0.1281        0.339        0.860
##    12      6       1    0.450  0.1347        0.250        0.809
##
##                 group=Placebo
##  time n.risk n.event survival std.err lower 95% CI upper 95% CI
##     1     21       4   0.8095  0.0857      0.65785        0.996
##     2     17       3   0.6667  0.1029      0.49268        0.902
##     3     14       2   0.5714  0.1080      0.39455        0.828
##     4     12       4   0.3810  0.1060      0.22085        0.657
##     6      8       4   0.1905  0.0857      0.07887        0.460
##     8      4       1   0.1429  0.0764      0.05011        0.407
##     9      3       1   0.0952  0.0641      0.02549        0.356
##    11      2       1   0.0476  0.0465      0.00703        0.322
##    12      1       1   0.0000     NaN           NA           NA
```

```
# Extract median (50th percentile) survival time by group
quantile(km_fit, probs = 0.5)
```

```
## $quantile
##               50
## group=6 MP    12
## group=Placebo  4
##
## $lower
##               50
## group=6 MP     8
## group=Placebo  2
##
## $upper
##               50
## group=6 MP    NA
## group=Placebo  6
```

The median survival time is greater for 6-MP, indicating patients on 6-MP stay in remission longer on average.

```
#c)
# Perform log-rank test
logrank_test <- survdiff(surv_obj ~ group, data = data)
print(logrank_test)
```

```
## Call:
## survdiff(formula = surv_obj ~ group, data = data)
##
##                N Observed Expected (O-E)^2/E (O-E)^2/V
## group=6 MP    21        9       19      5.25      16.7
## group=Placebo 21       21       11      9.06      16.7
##
##  Chisq= 16.7  on 1 degrees of freedom, p= 4e-05
```

It indicates a statistically significant difference (low p-value), thus 6-MP improves remission duration compared to placebo under this model.